

- 3 The photograph shows a caterpillar and fluid inside a pitcher plant found on the island of Madagascar.

The fluid contains extracellular enzymes which are digesting the caterpillar.



(Source: © Martin Lindsay/Alamy Stock Photo)

- (a) Describe the role of the Golgi apparatus in the formation of these extracellular enzymes.

(2)

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(b) The digested insects are a source of nitrogen for the pitcher plant.

Give **two** ways in which this source of nitrogen is used by the pitcher plants.

(2)

1

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(c) Bacterial cells have also been found in the fluid in pitcher plants.

(i) Which **three** structures are found in both plant cells and bacterial cells?

(1)

- ☐ **A** cell membrane, cellulose cell wall and DNA
- ☐ **B** cell membrane, DNA and pili
- ☐ **C** cell wall, DNA and capsule
- ☐ **D** cell wall, DNA and ribosome

(ii) The fluid of the pitcher plant contains water and molecules from digested organisms, which bacterial cells need for growth.

Explain **two** other conditions needed for optimum bacterial growth in the fluid of the pitcher plant.

(2)

1

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2

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(Total for Question 3 = 7 marks)

